**AKSHITH KALVAKOTA**

Akshithkalvakota123@gmail.com

AI/ML INTERN

# DAY-3

## TASK 2 : UNDERSTANDING LOGIC BEHIND BODY MEASUREMENTS

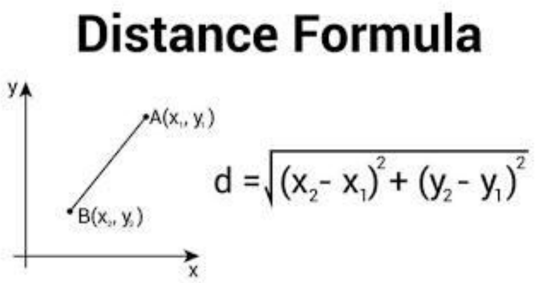
Body measurements is the core of the Raritone app, if it can't calculate accurate body measurements it will fail to suggest correct sizes for the customers in the Virtual Try-On feature.

MediaPipe does not directly give body measurements. It gives body landmarks in the form of coordinates(x,y) and then we have to calculate measurements using the distance between landmarks. It uses three major body measurement estimations namely:

1. Shoulder width estimation
2. Height Estimation
3. Waist estimation

### SHOULDER-WIDTH ESTIMATION:

To estimate this, Each side of the shoulder (left and right) is assigned a landmark with an index number. Then the distance between these indexes in measured using their coordinates using formulae like Euclidean distance.




### WORKFLOW:

Webcam/Image -> OpenCV captures frame -> MediaPipe detects body landmarks -> Get left shoulder and right shoulder -> Calculate distance between them -> Estimated Shoulder Width

### HEIGHT ESTIMATION:

To estimate this, Head to toes or Nose to toes measurement is taken with their assigned indexes. Then we use Euclidean Formula to get the distance.

## WORKFLOW:

Webcam/Image → OpenCV → MediaPipe Pose Detection → Head & Foot Landmarks → Distance Calculation → Estimated Height

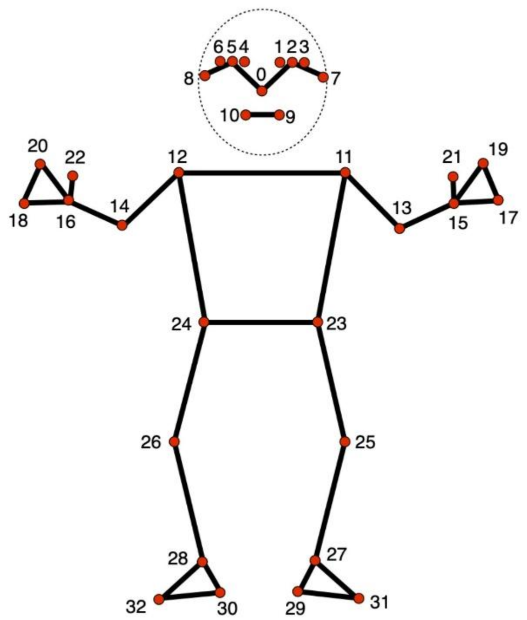
## WAIST ESTIMATION:

To estimate this, we take the distance between the hip landmarks namely left hip and right hip landmark.

## WORKFLOW:

Webcam/Image → OpenCV → MediaPipe Pose Detection → Hip Landmarks → Distance Calculation → Estimated Waist Width

LET'S TAKE A DIAGRAM TO UNDERSTAND:




This diagram has assigned indexes for all 33 body landmarks lets see which indexed we consider to calculate what.

For Shoulder Width, we take distance between 11 and 12.

For Height Estimation, we take distance between 0 and 29 or 30.

For Waist Estimation, we take distance between 23 and 24.

**THINGS TO CONSIDER:**

1. Real measurements are not given (cm/m/inches) instead it gives distance based on pixels.
2. The perspective of the camera could affect measurements. If person is too close, he/she could look wider than usual. If person is too far, he/she could look thinner than usual.
3. Loose clothing could affect the measurements.